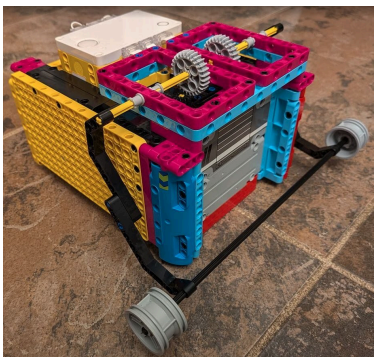
Rosie Robotics, Team #75872: Robot Design

Rookie team · 2026 FLL Challenge: Bioglow · Robot: Rosie Rover · Coded in Pybricks (Blocks to Python)

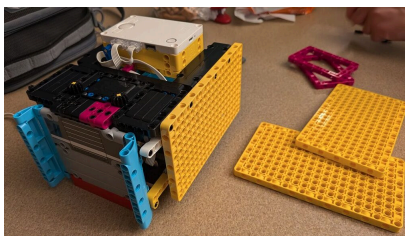




Rubric	Who's talking	What we'll show
Identify	**Bowen**, **Lina**	Resources used, robot base, Pybricks + remote, mission strategy
Design	**Miriam**, **James**	Mission 1 Drone Survey plan, motor-driven arm (Mission 2 Exploding Seeds, Mission 7 Humongous Fungus)
Create	**Jennie**	Mission 13 Keystone Species code and attachment
Iterate	**Neala**	How testing changed Mission 13 Keystone Species
Communicate	**Nolan**, **Myra**	Lessons learned

Identify: Resources & the Robot Base (Bowen)

- **As a rookie team, we started by learning from veteran teams.** The YouTube video *"5 Simple Tips to Build the Ultimate FLL Robot"* recommends building one strong **base robot** first, then small attachments for each mission.
- **Base:** built following the video's tips, then finished with an original design for the outside panels. It has 2 drive motors, **2 attachment motors on top**, and 2 color sensors. We made a Pybricks template program for this setup.

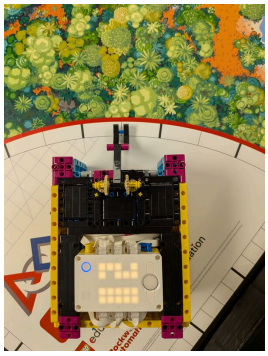




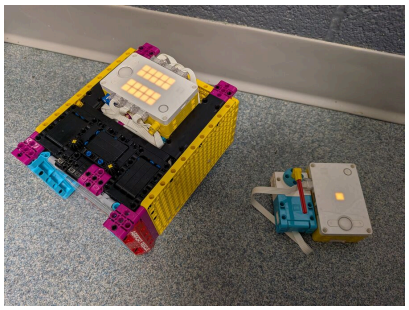
- **Pybricks instead of the regular SPIKE app:** missions start in Pybricks Blocks, then move to Python. Each mission is its own file, and all the code is on GitHub (https://github.com/Rosie-Rover-FLL/bioglow_2026).



- **Mission menu on the hub:** press left/right to pick a mission number, center to run it.

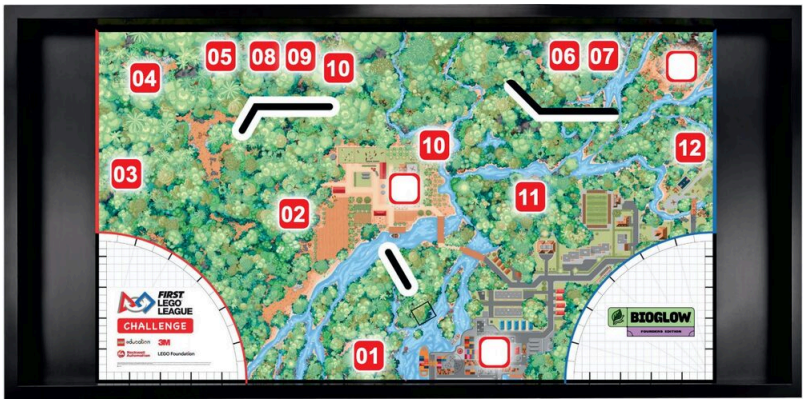


- **Custom remote control:** a second hub. Tilt it to drive, a knob sets the speed, and holding a button moves the arm. Used for driving practice and testing attachments. (live demo)



Identify: Mission Strategy (Lina)

Missions chosen for **reliability** and the **easiest points**: close to a launch area, with simple moves.



Mission	Goal pts
Equipment inspection	20
01 Drone Survey	20
02 Exploding Seeds	20
03 Flip the Rock	30
06 Leafcutter Frenzy (also scores 11 Window to the Past)	50
07 Humongous Fungus	20
10 Fragile Microhabitats (don't touch it!)	20
11 Window to the Past	20
13 Keystone Species	30
15 Biocentric Architecture	10
Precision tokens	50
Goal	**290**

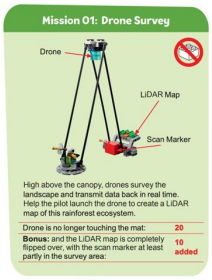
Design: Mission 1 Drone Survey and the Arm (Miriam, James)

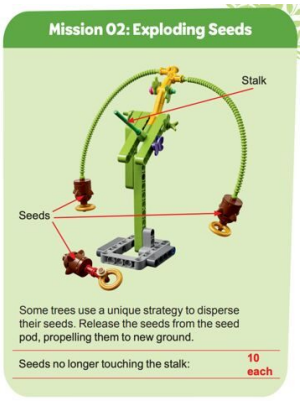
Mission 1 Drone Survey (Miriam): the robot drives backward fast to knock the drone off the mat, then drives straight home.

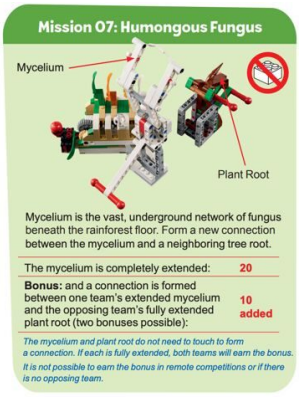
The arm, Mission 2 Exploding Seeds (James): both top motors turn the arm together. The first version was too weak, so the speed went way up. The arm lowers again on the way home, ready for the next run. We might even be able to use the same arm for **Mission 7 Humongous Fungus**.



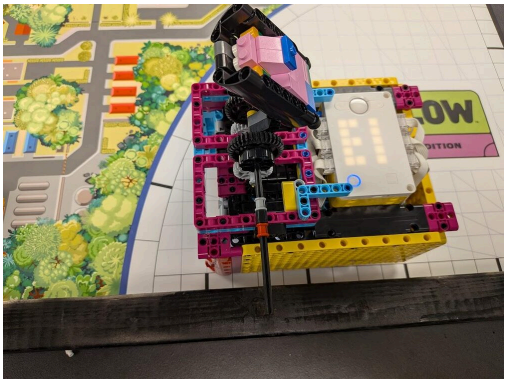








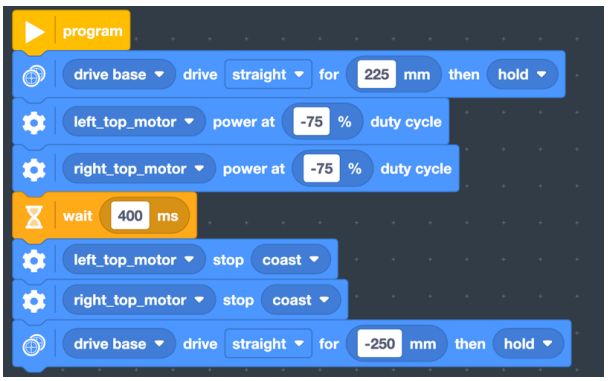
Create: Mission 13 Keystone Species (Jennie)





Attachment: connects to the top motors with gears, and throws the keystone species onto the restoration platform.

Code (Jennie): first written in Pybricks Blocks. Drive to the platform, run both top motors at 75% power for 0.4 seconds, then back up.



Iterate: How Testing Changed Mission 13 Keystone Species (Neala)

Repeated test runs showed two problems:

Problem found in testing	Fix
Not enough power to raise the young trees	Added **weight** to the keystone species, and raised the throw from 75% to **100% power**
The species sometimes missed the center of the platform	Drive out **slower** (300 mm/s) to stop wheel slip, but still drive back **fast** (600 mm/s)



These edits were made directly in the Python code (changes marked ◀):

```
drive_base.settings(straight_speed=300)   # ◀ NEW: slow drive out, no wheel
slip
drive_base.straight(225)
left_top_motor.dc(-100)                   # ◀ was -75: full-power throw
right_top_motor.dc(-100)                  # ◀ was -75
wait(400)
left_top_motor.stop()
right_top_motor.stop()
drive_base.settings(straight_speed=600)   # ◀ NEW: fast drive home
drive_base.straight(-250)
```

Communicate: Lessons Learned (Nolan, Myra)

1. **A robot doesn't do the same thing every time.** The battery, where the robot is placed, and wheel slip all matter. The fixes: stall-detect code (if the robot gets stuck, it stops and moves on), and gentler starts so the wheels don't slip.
2. **Precision tokens are worth protecting.** Each touch of the robot outside home costs a token. Sometimes it's smarter to let a mission fail.
3. **Practice runs taught us the scoring.**

	Score
First full practice run (Oct 11)	180
...many more runs not shown...	...
Practice run (Nov 1)	230
Practice run (Nov 8)	220
Current goal	**290**

The score sheet used at every practice run:

Mission	Got	Goal	Max
Equipment inspection All equipment in one launch area, under 12 in.		20	20
01 Drone Survey Drone off mat 20 · LiDAR bonus 10		20	30
02 Exploding Seeds 10 per seed off the stalk (3)		20	30
03 Flip the Rock Flag down 20 · rock returned 10		30	30
04 Lucky Leaves One leaf 10 · second leaf bonus 20		0	30
05 Reaching Roots Partial 10 or complete 20		0	20
06 Leafcutter Frenzy 10 per fragment + ant in the nest		50	50
07 Humongous Fungus Mycelium extended 20 · connection bonus 10		20	30
08 Tangled Vine touching the mat		0	30
09 Research Platform Platform 10 · camera trap 10 · seed 10		0	30
10 Fragile Microhabitats Spider 10 and Snail 10 left in place		20	20
11 Window to the Past Root cover down, touching the mat		20	20
12 Forest Elder Cane raised 20 · tie around post 10		0	30
13 Keystone Species Species on platform, young trees raised		30	30
14 Seeds of Renewal 5 per seed in station · +5 touching mat		0	40
15 Biocentric Architecture Canopy 10 Skylight 10 Hatch 10 Dock bonus 10		10	40
Precision tokens Points for tokens left at the end		50	50
Total	**0**	**290**	**530**

Proud of Rosie Rover! As a brand-new team, we used **Discovery, Innovation, Impact, Inclusion, Teamwork, and Fun** to build it together.

Thank you, judges!